United States Patent 12413000
Kind Code B2
Date of Patent September 09, 2025
Inventor(s) Ikeda; Ryota

Flat conductor electric connector

Abstract

A flat conductor electric connector includes: a housing with a receiving space open to a back such that the flat conductor is inserted forward into the housing; and multiple terminals arrayed and held on the housing with plate surfaces of the terminals at right angle to a terminal array direction, wherein each terminal has an arm portion with a support target arm portion on an inner surface of a wall portion extending in a front-back direction, and with an arm portion having a first portion extending from a back end of the support target arm portion and elastically displaceable in a thickness direction of the flat conductor, and a second portion, on a receiving space side from the first elastic portion, being bent forward from a back end of the first portion, being elastically displaceable in the thickness direction, and having a contact portion contactable with the flat conductor.

Inventors: **Ikeda; Ryota (Kanagawa, JP)**

Applicant: **HIROSE ELECTRIC CO., LTD.** (Kanagawa, JP)

Family ID: **88696900**

Assignee: **HIROSE ELECTRIC CO., LTD. (Kanagawa, JP)**

Appl. No.: **18/319278**

Filed: **May 17, 2023**

Prior Publication Data

Document Identifier	Publication Date
US 20230378672 A1	Nov. 23, 2023

Foreign Application Priority Data

JP	2022-082144	May. 19, 2022

Publication Classification

Int. Cl.: **H01R12/00** (20060101); **H01R12/70** (20110101); **H01R12/72** (20110101); H01R13/627 (20060101)

U.S. Cl.:

CPC **H01R12/7058** (20130101); **H01R12/727** (20130101); H01R13/6272 (20130101)

Field of Classification Search

CPC: H01R (12/727); H01R (12/7058)

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
9876297	12/2017	Takane	N/A	H01R 13/635
10566721	12/2019	Nakajima	N/A	H01R 12/89
2006/0178039	12/2005	Chen	439/495	H01R 43/20
2017/0170585	12/2016	Yukutake	N/A	H01R 12/721
2019/0044263	12/2018	Manba	N/A	H01R 12/88
2019/0190167	12/2018	Choi et al.	N/A	N/A
2020/0251842	12/2019	Takane et al.	N/A	N/A
2021/0359449	12/2020	Manba	N/A	N/A

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
109075483	12/2017	CN	N/A
109997281	12/2018	CN	N/A
2019-67717	12/2018	JP	N/A

OTHER PUBLICATIONS

Taiwanese Office Action (TWOA) issued on Jun. 10, 2025 for Taiwanese Patent Application No. 112115973 and its English machine translation. cited by applicant

---

Primary Examiner: Nguyen; Phuong Chi Thi

Attorney, Agent or Firm: RANKIN, HILL & CLARK LLP

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION
(1) This application claims priority from Japanese Patent Application No. 2022-082144 filed with the Japan Patent Office on May 19, 2022, the entire content of which is hereby incorporated by reference.
BACKGROUND

1. Technical Field
(2) The present disclosure relates to a flat conductor electric connector to which a flat conductor is connected.
2. Related Art
(3) A connector to which a band-shaped flat conductor extending in a front-back direction and having a thickness in an up-down direction is insertably connected forward is disclosed in JP-A-2019-067717. JP-A-2019-067717 describes a backward direction as a flat conductor insertion direction and describes a forward direction as a flat conductor removal direction, but the forward direction will be described herein as the flat conductor insertion direction and the backward direction will be described herein as the flat conductor removal direction. The connector of JP-A-2019-067717 is mounted on a mount surface of a circuit board, and multiple terminals arrayed with a flat conductor band width direction as a terminal array direction are held on a housing. Moreover, a lock member for preventing detachment of a flat conductor is turnably supported by the housing.
(4) The terminal is formed in such a manner that a metal plate member is punched into a flat plate shape, and is provided in such a posture that a plate surface thereof is at a right angle to the terminal array direction. The terminal has two contact pieces extending backward along a lower wall of the housing and elastically displaceable in the up-down direction, and at a contact provided at a back end of each contact piece, contacts the flat conductor from below.
SUMMARY
(5) A flat conductor electric connector according to an embodiment of the present disclosure is a flat conductor electric connector to which a flat conductor extending in a front-back direction is connected, including: a housing formed with a receiving space which is opened to a back such that the flat conductor is inserted forward into the housing; and multiple terminals arrayed and held on the housing in a state in which plate surfaces of the terminals are at a right angle to a terminal array direction which is a direction at a right angle to both the front-back direction and a thickness direction of the flat conductor, in which each terminal has an arm portion extending in the front-back direction, the arm portion has a support target arm portion supported on an inner surface of a wall portion and extending in the front-back direction, and an elastic arm portion extending backward from the support target arm portion, the elastic arm portion has a first elastic portion extending from a back end of the support target arm portion and elastically displaceable in the thickness direction of the flat conductor without the support target arm portion supported on the wall portion, and a second elastic portion positioned on a receiving space side with respect to the first elastic portion, bent forward from a back end of the first elastic portion, and elastically displaceable in the thickness direction of the flat conductor, and the second elastic portion has a contact portion contactable with the flat conductor with contact pressure.

Description

BRIEF DESCRIPTION OF DRAWINGS
(1) FIG. **1** is a perspective view showing, together with a flat conductor, a flat conductor electric connector according to an embodiment of the present invention, and shows a state immediately before the flat conductor is connected;
(2) FIG. **2** is a perspective view showing, together with the flat conductor, the flat conductor electric connector according to the embodiment of the present invention, and shows a state immediately before the flat conductor is removed;
(3) FIG. **3** is a perspective view showing a state in which terminals, metal fittings, and a movable member are disassembled from the flat conductor electric connector;
(4) FIG. **4** is a longitudinal sectional view of the flat conductor electric connector, and shows the section at the position of the terminal in a connector width direction;

(5) FIGS. **5**A and **5**B are perspective sectional views of the flat conductor electric connector and show the longitudinal section at the position of the metal fitting in the connector width direction, FIG. **5**A showing a state in which one metal fitting is removed and FIG. **5**B showing a state in which the metal fitting is attached;
(6) FIGS. **6**A to **6**C are longitudinal sectional views of the flat conductor electric connector in a state immediately before the flat conductor is inserted, FIG. **6**A showing the section at the position of the terminal, FIG. **6**B showing the section at the position of a locking portion of the movable member, and FIG. **6**C showing the section at the position of a cam portion of the movable member;
(7) FIGS. **7**A to **7**C are longitudinal sectional views of the flat conductor electric connector in a state in which insertion of the flat conductor is completed, FIG. **7**A showing the section at the position of the terminal, FIG. **7**B showing the section at the position of the locking portion of the movable member, and FIG. **7**C showing the section at the position of the cam portion of the movable member; and
(8) FIGS. **8**A to **8**C are longitudinal sectional views of the flat conductor electric connector in a state immediately before the flat conductor is removed, FIG. **8**A showing the section at the position of the terminal, FIG. **8**B showing the section at the position of the locking portion of the movable member, and FIG. **8**C showing the section at the position of the cam portion of the movable member.

DETAILED DESCRIPTION

(9) In the following detailed description, for purpose of explanation, numerous specific details are set forth in order to provide a thorough understanding of the disclosed embodiments. It will be apparent, however, that one or more embodiments may be practiced without these specific details. In other instances, well-known structures and devices are schematically shown in order to simplify the drawing.
(10) In JP-A-2019-067717, the contact piece of the terminal extends backward, and the contact is provided at the back end of the contact piece. Thus, there is a probability that the contact piece is buckled due to collision of a front end of the flat conductor with the contact of the contact piece from the back upon insertion connection of the flat conductor. Due to buckling of the contact piece, the contact piece and the flat conductor cannot properly contact each other.
(11) The present invention has been made in view of the above-described situation, and is intended to provide a flat conductor electric connector configured such that buckling of a terminal when a flat conductor is inserted into and connected to the connector can be avoided and the terminal and the flat conductor can properly contact each other.
(12) (1) A flat conductor electric connector according to the present invention is a flat conductor electric connector to which a flat conductor extending in a front-back direction is connected, including: a housing formed with a receiving space which is opened to a back such that the flat conductor is inserted forward into the housing; and multiple terminals arrayed and held on the housing in a state in which plate surfaces of the terminals are at a right angle to a terminal array direction which is a direction at a right angle to both the front-back direction and a thickness direction of the flat conductor.
(13) In the flat conductor electric connector, in the present invention, each terminal has an arm portion extending in the front-back direction, the arm portion has a support target arm portion supported on an inner surface of a wall portion and extending in the front-back direction, and an elastic arm portion extending backward from the support target arm portion, the elastic arm portion has a first elastic portion extending from a back end of the support target arm portion and elastically displaceable in the thickness direction of the flat conductor without the support target arm portion supported on the wall portion, and a second elastic portion positioned on a receiving space side with respect to the first elastic portion, bent forward from a back end of the first elastic portion, and elastically displaceable in the thickness direction of the flat conductor, and the second elastic portion has a contact portion contactable with the flat conductor with contact pressure.

(14) In the present invention, the second elastic portion of the terminal having the contact portion is bent at the back end of the first elastic portion, and extends forward. Thus, when the flat conductor is inserted into the receiving space of the housing from the back, the flat conductor is smoothly guided forward by back end portions of the terminals, i.e., bent portions of the elastic arm portions curved in a backwardly-raised shape, and reaches the position of the contact portions of the second elastic portions. As a result, at any portion, the terminal is not buckled due to contact with a front end of the flat conductor. Thus, proper contact pressure between the terminals and the flat conductor can be easily ensured.
(15) (2) In the aspect (1) of the invention, the first elastic portion may be formed thicker and longer than the second elastic portion as viewed in the terminal array direction, and a front end of the first elastic portion may be positioned at a front with respect to a front end of the second elastic portion.
(16) Generally, in order to avoid concentration of a load on part of the elastic arm portion due to contact with the flat conductor in a case where the elastic arm portion for contact with the flat conductor is provided at the terminal, a portion of the elastic arm portion on a free end side is formed thin, and the opposite portion thereof is formed thick in many cases. With this configuration, stress on the elastic arm portion is dispersed. In the aspect (2) of the invention, in the elastic arm portion, the first elastic portion is formed thicker, and the second elastic portion positioned on the free end side is formed thinner than the first elastic portion. With this configuration, the stress is dispersed. If the thick first elastic portion and the thin second elastic portion are formed with the same length, it is difficult to ensure proper contact pressure due to a great difference in spring properties between both these elastic portions when the contact portion of the second elastic portion contacts the flat conductor. On the other hand, in the aspect (2) of the invention, the first elastic portion thicker than the second elastic portion is formed longer, and therefore, the spring properties can be set to a similar degree between the first elastic portion and the second elastic portion, and as a result, a proper pressure of contact with the flat conductor is easily ensured.
(17) (3) In the aspect (1) or (2) of the invention, each terminal may have another arm portion positioned on the opposite side of the receiving space from the arm portion and extending in the front-back direction, the other arm portion may have another contact portion contacting the flat conductor with contact pressure to sandwich the flat conductor in cooperation with the contact portion, and a distance between the contact portion and the other contact portion in the front-back direction may be smaller than the thickness dimension of the flat conductor.
(18) Since the distance between the contact portion and the other contact portion in the front-back direction is set smaller than the thickness dimension of the flat conductor as described above, even when the flat conductor tilts due to external force applied in the thickness direction of the flat conductor in a state in which the flat conductor is connected to the connector, a state in which the flat conductor is sandwiched by the contact portion and the other contact portion is easily maintained. As a result, contact between the flat conductor and each terminal with proper contact pressure can be ensured, and unexpected detachment of the flat conductor can be avoided.
(19) (4) In the aspect (1) or (2) of the invention, each terminal may have another arm portion positioned on the opposite side of the receiving space from the arm portion and extending in the front-back direction, the other arm portion may have another contact portion contacting the flat conductor with contact pressure to sandwich the flat conductor in cooperation with the contact portion, and a distance between the contact portion and the other contact portion in the front-back direction may be smaller than twice as great as a clearance dimension between the contact portion and the other contact portion in the thickness direction of the flat conductor.
(20) Generally, in a case where the flat conductor is sandwiched by the contact portion and the other contact portion, the clearance dimension between the contact portion and the other contact portion in the thickness direction of the flat conductor is often set smaller than the half of the thickness dimension of the flat conductor in order to ensure sufficient contact pressure for the flat

conductor. Thus, the distance between the contact portion and the other contact portion in the front-back direction is set smaller than twice as great as the above-described clearance dimension. With this configuration, even when the flat conductor tilts due to the external force applied in the thickness direction of the flat conductor in the state in which the flat conductor is connected to the connector, the state in which the flat conductor is sandwiched by the contact portion and the other contact portion is easily maintained. As a result, contact between the flat conductor and each terminal with proper contact pressure can be ensured, and unexpected detachment of the flat conductor can be avoided.
(21) In the present invention, buckling of the terminal when the flat conductor is inserted into and connected to the connector can be avoided, and the terminal and the flat conductor can properly contact each other.
(22) Hereinafter, an embodiment of the present invention will be described based on the attached drawings.
(23) FIGS. **1** and **2** are perspective views showing, together with a flat conductor C, a flat conductor electric connector **1** (hereinafter referred to as a “connector **1**”) according to the present embodiment, FIG. **1** showing a state immediately before the flat conductor C is connected and FIG. **2** showing a state immediately before the flat conductor C is removed. FIG. **3** is a perspective view showing the connector **1** with terminals **20**, metal fittings **30**, and a movable member **40** disassembled from the connector **1**.
(24) The connector **1** is mounted on a mount surface of a circuit board (not shown), and the flat conductor C (e.g., an FPC) which is a partner connector is connected so as to be insertable into or removable from the connector **1** with a front-back direction (X-axis direction) parallel with the mount surface as an insertion-removal direction. The connector **1** brings the circuit board and the flat conductor C into electrical conduction with each other in such a manner that the flat conductor C is connected to the connector **1**. In the present embodiment, in the X-axis direction (front-back direction), an X1 direction is a forward direction, and an X2 direction is a backward direction. Moreover, a Y-axis direction at a right angle to the front-back direction (X-axis direction) is a connector width direction, and a Z-axis direction at a right angle to the mount surface of the circuit board is an up-down direction.
(25) The flat conductor C is in a flexible band shape which extends in the front-back direction (X-axis direction) and of which the width direction is the connector width direction (Y-axis direction) and thickness direction is the up-down direction (Z-axis direction), and as shown in FIG. **2**, a front end portion of the flat conductor C is inserted into a housing **10**. The flat conductor C is formed such that multiple circuit portions (not shown) extending in the front-back direction are arrayed in the connector width direction. The circuit portions extend, with embedded in an insulating layer of the flat conductor C, in the front-back direction to a position in the vicinity of a front end of the flat conductor C. Moreover, the circuit portions are exposed at a lower surface of the front end portion, and are connectable to the later-described terminals **20** of the connector **1**.
(26) As shown in FIG. **1**, the front end portion of the flat conductor C is narrower than other portions. Cutouts C1 are formed at both side edges of the front end portion, and back edges of ear portions C2 positioned at the front of the cutouts C1 function as locking target portions C2A to be locked to later-described locking portions **42** of the connector **1** (see FIG. **7**B). At front end portions of the other portions of the flat conductor C, shoulder portions C3 are formed on both sides in a width direction of the flat conductor C. As shown in FIG. **2**, the shoulder portions C3 are positioned close to later-described back end wall portions **18** of the housing **10** at the back thereof in a state in which the flat conductor C is inserted into the connector **1**.
(27) As shown in FIGS. **1** to **3**, the connector **1** includes the housing **10** made of an electric insulating material such as resin, the multiple terminals **20** (also see FIG. **4**) formed of metal plates, arrayed with the connector width direction as a terminal array direction, and held on the housing **10**, the metal fittings **30** formed of metal plates and arranged on both outer sides of a terminal array

area in the connector width direction, and the movable member **40** provided turnable between a closed position and an open position and made of an electric insulating material such as resin or metal. The flat conductor C is inserted into and connected to the connector **1** from the back (see an arrow shown in FIG. **1**).
(28) As shown in FIGS. **1** and **2**, the housing **10** has a substantially rectangular parallelepiped outer shape with the connector width direction as a longitudinal direction, and a receiving space **11** for receiving the flat conductor C is formed as a space opened to the back. The housing **10** has a lower wall **12** and an upper wall **13** which are wall portions extending parallel with the mount surface of the circuit board, two side walls **14** (see FIG. **3**) extending in the up-down direction and coupling both end portions of the lower wall **12** in the connector width direction and both end portions of the upper wall **13** in the connector width direction to each other, and a front wall **15** (see FIG. **6**B) coupling front ends of the lower wall **12** and the upper wall **13** to each other.
(29) As shown in FIG. **3**, the housing **10** has, outside the side walls **14** in the connector width direction, projecting portions **16** projecting from outer surfaces of lower portions of the side walls **14**, metal fitting holding portions **17** provided at front half portions of the projecting portions **16**, and the back end wall portions **18** provided at back end portions of the projecting portions **16**.
(30) The receiving space **11** is surrounded by the lower wall **12**, the upper wall **13**, the front wall **15** (see FIG. **6**B), and the two side walls **14** (see FIG. **3**), extends in the front-back direction, and receives the front end portion of the flat conductor C (see FIGS. **7**A and **7**B).
(31) At the upper wall **13**, upper holes **13**A penetrating the upper wall **13** in the up-down direction outside the terminal array area are formed. As shown in FIG. **1**, the upper holes **13**A allow the later-described locking portions **42** of the movable member **40** to enter the upper holes **13**A from above when the movable member **40** is at the closed position (also see FIG. **6**B).
(32) As shown in FIGS. **1** to **3**, in the housing **10**, terminal housing portions **19** in which the terminals **20** are housed are formed and arrayed in the connector width direction. FIG. **4** is a longitudinal sectional view of the connector **1** along a plane at a right angle to the connector width direction, and shows the section at the position of the terminal **20** in the connector width direction. As shown in FIG. **4**, the terminal housing portion **19** is formed as a groove portion penetrating the housing **10** in the front-back direction. The terminal housing portion **19** has a lower groove portion **19**A recessed from an upper surface of the lower wall **12** and extending in the front-back direction, an upper groove portion **19**B recessed from a lower surface of the upper wall **13** and extending in the front-back direction, and a front groove portion **19**C penetrating the front wall **15** in the front-back direction, extending in the up-down direction, and coupling front end portions of the lower groove portion **19**A and the upper groove portion **19**B to each other. The terminal housing portion **19** is, as a whole, in a backwards C-shape opened to the back.
(33) As shown in FIG. **3**, the projecting portion **16** is in a plate shape having a plate surface at a right angle to the up-down direction, and in the front-back direction, extends across an area over the entirety of the side wall **14**. The metal fitting holding portion **17** is provided with a clearance from an outer surface of the side wall **14** in the connector width direction. A space formed between the metal fitting holding portion **17** and the side wall **14** forms a side plate housing portion **10**A for housing a later-described side plate portion **43** of the movable member **40**.
(34) The metal fitting holding portion **17** is in a standing plate shape having a plate surface at a right angle to the connector width direction, and a metal fitting housing portion **17**A for housing part of the metal fitting **30** is formed outside the projecting portion **16** in the connector width direction (also see FIGS. **5**A and **5**B). As shown in FIG. **3**, a portion of the metal fitting holding portion **17** positioned inside the metal fitting housing portion **17**A in the connector width direction stands upward from the projecting portion **16**. A lower portion of the metal fitting holding portion **17** is provided with a locking target end portion **17**B which protrudes backward with respect to a back end of the metal fitting housing portion **17**A. The locking target end portion **17**B is positioned in an area including the metal fitting housing portion **17**A in the connector width direction, and is

positioned within the area of the projecting portion **16** in the up-down direction and is locked to a later-described locking leg portion **34** of the metal fitting **30** in the up-down direction (see FIG. **5**B).

(35) FIGS. **5**A and **5**B are longitudinal sectional views of the connector **1**, and shows the section at the position of the metal fitting **30** in the connector width direction. FIG. **5**A shows a state in which one metal fitting **30** is removed, and FIG. **5**B shows a state in which the metal fitting **30** is attached. As shown in FIGS. **5**A and **5**B, the metal fitting housing portion **17**A is in a groove shape extending at a right angle to the connector width direction. A lower portion of the metal fitting housing portion **17**A extends forward with respect to other portions, and as shown in FIG. **5**B, a later-described press-fitting fixing portion **31**A-**1** of the metal fitting **30** is held in a press-fitting groove portion **17**A-**1** formed on the front end side of such a lower portion.

(36) As shown in FIG. **3**, the back end wall portion **18** is positioned at the back of the metal fitting holding portion **17**, stands upward from the projecting portion **16**, and is coupled to the outer surface of the side wall **14**. A space formed between the metal fitting holding portion **17** and the back end wall portion **18** in the front-back direction forms a cam housing portion **10**B for housing a later-described cam portion **44** of the movable member **40**.

(37) The terminal **20** is formed by punching of a metal plate member, and in a state in which a plate surface thereof is at a right angle to the connector width direction, is press-fitted and attached from the front to the terminal housing portion **19** of the housing **10**. As shown in FIG. **4**, the terminal **20** has a lower arm portion **21** which is an arm portion extending in the front-back direction along the lower wall **12** of the housing **10**, an upper arm portion **22** which is another arm portion extending in the front-back direction along the upper wall **13** of the housing **10**, a coupling arm portion **23** extending in the up-down direction along the front wall **15** and coupling front end portions of the lower arm portion **21** and the upper arm portion **22** to each other, and a connection portion **24** extending forward from a lower portion of the coupling arm portion **23**.

(38) The lower arm portion **21** has a support target arm portion **21**A housed in the lower groove portion **19**A, supported from below by a groove bottom surface **19**A-**1** forming an inner surface of the lower groove portion **19**A, and extending in the front-back direction, and an elastic arm portion **21**B extending backward from the support target arm portion **21**A. The support target arm portion **21**A is entirely housed in the lower groove portion **19**A. An upper edge of the support target arm portion **21**A is inclined downward as extending backward, and on the other hand, a lower edge of the support target arm portion **21**A is not inclined with respect to the front-back direction. That is, the support target arm portion **21**A is gradually narrowed toward the back. The lower edge of the support target arm portion **21**A is supported on the groove bottom surface **19**A-**1** across the entire area of the support target arm portion **21**A in the front-back direction.

(39) The elastic arm portion **21**B has a first elastic portion **21**B-**1** extending from a back end of the support target arm portion **21**A, and a second elastic portion **21**B-**2** positioned higher than the first elastic portion **21**B-**1**, i.e., on a receiving space **11** side, and bent forward from a back end of the first elastic portion **21**B-**1**. The first elastic portion **21**B-**1** is entirely housed in the lower groove portion **19**A, and is inclined upward as extending backward. That is, a clearance formed between a lower edge of the first elastic portion **21**B-**1** and the groove bottom surface **19**A-**1** of the lower groove portion **19**A gradually becomes larger toward the back. The first elastic portion **21**B-**1** is elastically deformable in the up-down direction within the area of this clearance. Hereinafter, this clearance will be referred to as an “elastically-deformable space **19**A-**2**.”

(40) In the present embodiment, the groove bottom surface **19**A-**1** of the lower groove portion **19**A is formed at the same height across the entire area in the front-back direction, and therefore, the elastically-deformable space **19**A-**2** is formed as the clearance between the first elastic portion **21**B-**1** and the groove bottom surface **19**A-**1** because the first elastic portion **21**B-**1** is inclined. However, the form of the elastically-deformable space **19**A-**2** is not limited to above. For example, as a modification, in an area corresponding to the first elastic portion **21**B-**1** in the front-back

direction, the groove bottom surface **19**A-**1** may be formed lower than other areas, and accordingly, the elastically-deformable space **19**A-**2** may be formed. As another modification, a space below the first elastic portion **21**B-**1**, i.e., a downwardly-opened space, may be formed as the elastically-deformable space **19**A-**2** without the lower wall **12** formed in the area corresponding to the first elastic portion **21**B-**1** in the front-back direction. In these modifications, the first elastic portion **21**B-**1** may be in a shape extending without inclined with respect to the front-back direction.

(41) As shown in FIG. **4**, the second elastic portion **21**B-**2** is partially housed in the lower groove portion **19**A, and is inclined upward as extending backward. At a front end of the second elastic portion **21**B-**2**, a lower contact portion **21**B-**3** which is a contact portion contactable with the circuit portion at the lower surface of the flat conductor C with contact pressure is provided so as to protrude upward. When the second elastic portion **21**B-**2** is in a free state, the lower contact portion **21**B-**3** protrudes from the lower groove portion **19**A, and is positioned in the receiving space **11**. A clearance formed between the second elastic portion **21**B-**2** and the first elastic portion **21**B-**1** in the up-down direction gradually becomes larger toward the back, and in the area of this clearance, the second elastic portion **21**B-**2** is elastically displaceable in the up-down direction.

(42) As shown in FIG. **4**, the first elastic portion **21**B-**1** is formed thicker than the second elastic portion **21**B-**2**. Since the second elastic portion **21**B-**2** of the elastic arm portion **21**B positioned on a free end side is formed thin and the first elastic portion **21**B-**1** of the elastic arm portion **21**B positioned on the side opposite to a free end is formed thick as described above, stress on the elastic arm portion **21**B due to contact with the flat conductor C can be dispersed, and therefore, concentration of a load on part of the elastic arm portion **21**B can be avoided.

(43) As shown in FIG. **4**, the first elastic portion **21**B-**1** is formed longer than the second elastic portion **21**B-**2**, and a front end of the first elastic portion **21**B-**1** is positioned at the front with respect to the front end of the second elastic portion **21**B-**2**. Since the first elastic portion **21**B-**1** thicker than the second elastic portion **21**B-**2** is formed longer than the second elastic portion **21**B-**2** in the present embodiment as described above, spring properties can be set to a similar degree between the first elastic portion **21**B-**1** and the second elastic portion **21**B-**2**, and as a result, a proper pressure of contact with the flat conductor C is easily ensured.

(44) As shown in FIG. **4**, the upper arm portion **22** is positioned on the opposite side of the receiving space **11** from the lower arm portion **21** in the up-down direction, i.e., above the lower arm portion **21**, and the upper arm portion **22** is inclined downward as extending backward in a state in which part of the upper arm portion **22** is housed in the upper groove portion **19**B. At a back end of the upper arm portion **22**, an upper contact portion **22**A which is another contact portion contactable with an upper surface of the flat conductor C with contact pressure is provided so as to protrude downward. As shown in FIG. **4**, when the upper arm portion **22** is in a free state, the upper contact portion **22**A protrudes from the upper groove portion **19**B, and is positioned in the receiving space **11**.

(45) As shown in FIG. **4**, a protruding tip of the upper contact portion **22**A is positioned at the front with respect to a protruding tip of the lower contact portion **21**B-**3** of the lower arm portion **21** by a distance P. In order to ensure sufficient contact pressure for the flat conductor C, a clearance having a dimension Q smaller than the half of the thickness dimension R (see FIG. **6**A) of the flat conductor C is formed between the protruding tip of the upper contact portion **22**A and the protruding tip of the lower contact portion **21**B-**3** in the up-down direction. Moreover, in the present embodiment, the distance P is set smaller than the thickness dimension R of the flat conductor C and smaller than twice as great as the clearance dimension Q.

(46) Since the distance P is set smaller than the thickness dimension R of the flat conductor C as described above, a state in which the flat conductor C is sandwiched with contact pressure by cooperation of the lower contact portion **21**B-**3** and the upper contact portion **22**A is easily maintained even when the flat conductor C tilts due to external force applied in the thickness direction (up-down direction) of the flat conductor C in a state in which the flat conductor C is

connected to the connector **1**. As a result, contact between the flat conductor C and each terminal **20** with proper contact pressure can be ensured, and unexpected detachment of the flat conductor C can be avoided.

(47) Because of the clearance dimension Q being set smaller than the half of the thickness dimension R of the flat conductor C, even when the distance P is set smaller than twice as great as the clearance dimension Q, the state in which the flat conductor C is sandwiched with contact pressure by cooperation of the lower contact portion **21**B-**3** and the upper contact portion **22**A is easily maintained. Thus, even when the flat conductor tilts due to the external force applied in the thickness direction (up-down direction) of the flat conductor in a state in which the flat conductor C is connected to the connector, contact between the flat conductor C and each terminal **20** with proper contact pressure can be ensured, and unexpected detachment of the flat conductor C can be avoided.

(48) In the present embodiment, both a condition where the distance P is smaller than the thickness dimension R of the flat conductor C and a condition where the distance P is smaller than twice as great as the clearance dimension Q are satisfied, but both these conditions are not necessarily satisfied, and unexpected detachment of the flat conductor C can be avoided as long as any of these conditions is satisfied.

(49) In the present embodiment, the protruding tip of the upper contact portion **22**A is positioned at the front with respect to the protruding tip of the lower contact portion **21**B-**3** of the lower arm portion **21**, but a relative positional relationship between the protruding tips in the front-back direction is not limited to above as long as the distance P between the protruding tips satisfies at least one of the above-described two conditions. That is, the protruding tip of the upper contact portion **22**A may be positioned at the back with respect to the protruding tip of the lower contact portion **21**B-**3** of the lower arm portion **21**, or the protruding tips may be at the same position.

(50) The coupling arm portion **23** is housed in the front groove portion **19**C. A press-fitting protrusion **23**A protruding upward is provided at an upper edge of the coupling arm portion **23**, and bites into an inner surface of the upper wall **13** to hold the terminal **20** in the terminal housing portion **19**. The connection portion **24** extends outward of the housing **10** from the lower portion of the coupling arm portion **23** to the lower front side. A lower edge of the connection portion **24** is positioned slightly lower than a lower surface of the lower wall **12** of the housing **10**, and in a state in which the connector **1** is arranged on the mount surface of the circuit board (not shown), is soldered to a corresponding circuit portion (pad) on the mount surface.

(51) FIGS. **5**A and **5**B are the perspective sectional views of the connector **1**, and show the longitudinal section at the position of the metal fitting **30** in the connector width direction. FIG. **5**A shows the state in which the one metal fitting **30** is removed, and FIG. **5**B shows the state in which the metal fitting **30** is attached. The metal fitting **30** is formed in such a manner that a metal plate member is punched and partially bent in a plate thickness direction, and is press-fitted in and attached to the metal fitting housing portion **17**A of the housing **10** from the back. Since the metal fitting **30** can be press-fitted as described above, the metal fitting **30** can be easily attached to the housing **10**.

(52) As shown in FIGS. **5**A and **5**B, the metal fitting **30** has a body portion **31** having a plate surface at a right angle to the connector width direction, a biasing portion **32** for biasing the cam portion **44** of the movable member **40**, a fixing leg portion **33** to be soldered to the mount surface of the circuit board, and the locking leg portion **34** lockable to the housing **10** from below (also see FIG. **3**). The body portion **31** having the plate surface at the right angle to the connector width direction forms a large portion of the metal fitting **30**, and only the biasing portion **32** and the fixing leg portion **33** are bent in the connector width direction as described later. Thus, the entirety of the metal fitting **30** is compactly formed with a simple shape.

(53) The body portion **31** has a fixing arm portion **31**A positioned adjacent to the cam portion **44** of the movable member **40** on the outer side thereof in the connector width direction, extending

linearly in the front-back direction, and fixed to the housing **10**, a movable arm portion **31**B positioned higher than the fixing arm portion **31**A and extending in the front-back direction, and a coupling portion **31**C extending in the up-down direction and coupling front end portions of the fixing arm portion **31**A and the movable arm portion **31**B to each other. The press-fitting fixing portion **31**A-**1** of the metal fitting **30** provided at the front end portion of the fixing arm portion **31**A is press-fitted in the press-fitting groove portion **17**A-**1** of the metal fitting housing portion **17**A, and accordingly, the metal fitting **30** is held in the metal fitting housing portion **17**A. In a state in which the metal fitting **30** is housed in the metal fitting housing portion **17**A, the fixing arm portion **31**A of the metal fitting **30** is supported on a lower inner wall surface **17**A-**2** of the metal fitting housing portion **17**A.

(54) The movable arm portion **31**B extends in the front-back direction, and is elastically deformable in the up-down direction, i.e., a direction parallel with a plate surface (rolled surface) thereof. A front half portion of the movable arm portion **31**B is inclined downward as extending backward, and a back half portion of the movable arm portion **31**B linearly extends, without inclination, backward from a back end of the front half portion. Note that the shape of the movable arm portion is not limited to above and the movable arm portion may be, for example, formed such that the front half portion is inclined upward as extending backward and the back half portion linearly extends, without inclination, backward from the back end of the front half portion. The coupling portion **31**C is inclined forward, at a position closer to a front end of the fixing arm portion **31**A, as extending upward from an upper edge of the fixing arm portion **31**A, and couples a portion of the fixing arm portion **31**A closer to the front end thereof and a front end portion of the movable arm portion **31**B to each other.

(55) The biasing portion **32** is bent, at an intermediate position of the back half portion of the movable arm portion **31**B in the front-back direction, at an upper edge of the movable arm portion **31**B, and extends inward in the connector width direction, i.e., toward the cam portion **44** of the movable member **40**. The biasing portion **32** has a plate surface (rolled surface) at a right angle to the up-down direction, is positioned immediately above the cam portion **44**, and restricts upward movement of the cam portion **44** and therefore upward movement of the movable member **40** (see FIGS. **6**C, **7**C, and **8**C).

(56) In the present embodiment, there is nothing above the back half portion of the movable arm portion **31**B as shown in FIGS. **1**, **2**, and **5**B, but as a modification, e.g., a restriction portion positioned above a back end portion of the movable arm portion **31**B with a predetermined clearance from such a back end portion may be formed by part of the housing **10**. In this modification, the back end portion of the movable arm portion **31**B contacts the restriction portion from below when displaced upward by a predetermined amount, and further displacement thereof is restricted. With this configuration, excessive upward elastic deformation of the movable arm portion **31**B is restricted. Thus, detachment of the cam portion **44** and therefore detachment of the movable member **40** from the housing **10** can be more reliably prevented.

(57) The fixing leg portion **33** is bent, at a position closer to a back end of the fixing arm portion **31**A, at a lower edge of the fixing arm portion **31**A, and extends inward in the connector width direction. Since the fixing leg portion **33** is formed so as to extend inward in the connector width direction, i.e., extend to the same side as extension of the biasing portion **32**, as described above, an increase in the size of the metal fitting **30** in the connector width direction due to the fixing leg portion **33** can be avoided. The fixing leg portion **33** is positioned at the back with respect to the biasing portion **32** in the front-back direction, and extends across the substantially same area as that of the biasing portion **32** in the connector width direction. Moreover, a lower surface of the fixing leg portion **33** is positioned slightly lower than the lower surface of the lower wall **12** of the housing (see FIGS. **6**C, **7**C, and **8**C), and in a state in which the connector **1** is arranged on the mount surface of the circuit board (not shown), the fixing leg portion **33** is soldered and fixed to a corresponding portion (pad) on the mount surface.

(58) The locking leg portion **34** is positioned at the front with respect to the fixing leg portion **33**, and is provided within the area of the biasing portion **32** in the front-back direction. The locking leg portion **34** extends forward after having extended downward from a lower edge of the fixing leg portion **33**, and the entire shape thereof is in an L-shape. As shown in FIG. **5**B, the L-shaped locking leg portion **34** is fitted in the locking target end portion **17**B of the metal fitting holding portion **17** from the back, and accordingly, is locked to the locking target end portion **17**B from below.
(59) As shown in FIGS. **1** to **3**, the movable member **40** has a plate-shaped operation portion **41** extending across an area between the side walls **14** (see FIGS. **2** and **3**) in the connector width direction, the locking portions **42** protruding from a plate surface of the operation portion **41**, and the side plate portions **43** and the cam portions **44** positioned at both ends of the operation portion **41** in the connector width direction.
(60) The operation portion **41** is operated when the movable member **40** is moved (turned) between the closed position shown in FIG. **1** and the open position shown in FIG. **2**. The locking portions **42** are provided at positions corresponding to the locking target portions C2A of the flat conductor C in the connector width direction at a back portion of the operation portion **41** at the closed position (lower portion of the operation portion **41** at the open position). When the movable member **40** is at the closed position, the locking portions **42** protrude from a lower surface of the operation portion **41**, and are positioned in the upper holes **13**A and receiving space **11** of the housing **10** (see FIG. **6**B). Thus, the locking portions **42** are contactable with the flat conductor C (see FIG. **7**B). A portion of the locking portion **42** positioned in the receiving space **11** has, at a back surface thereof, a guide surface **42**A inclined downward as extending forward, and has, at a front surface thereof, a locking surface **42**B locked to the locking target portion C2A of the flat conductor C from the back (see FIGS. **6**B and **7**B). When the movable member **40** is at the open position, the locking portions **42** are positioned outside the housing **10**, and contact with the flat conductor C is cancelled (see FIGS. **8**A and **8**B).
(61) As shown in FIG. **3**, the side plate portion **43** is, at a position at each end of the operation portion **41** in the connector width direction, in a plate shape having a plate surface at a right angle to the connector width direction. The side plate portion **43** extends across an area from an intermediate position of the operation portion **41** to a position at the back with respect to the operation portion **41** in the front-back direction, and protrudes downward. That is, a protruding end portion **43**A which is a back end portion of the side plate portion **43** shown in FIG. **3** protrudes backward from a back end of the operation portion **41**. The side plate portion **43** is entirely housed in the side plate housing portion **10**A of the housing **10** when the movable member **40** is at the closed position (see FIG. **1**), and a portion of the side plate portion **43** other than the protruding end portion **43**A is positioned outside the side plate housing portion **10**A when the movable member **40** is at the open position (see FIG. **2**).
(62) The cam portion **44** protrudes in a substantially rectangular columnar shape from an outer surface of the protruding end portion **43**A of the side plate portion **43** in the connector width direction, and regardless of the position of the movable member **40**, is constantly housed in the cam housing portion **10**B of the housing **10**. The cam portion **44** is positioned immediately below the biasing portion **32** of the metal fitting **30**, and upward movement thereof is restricted by the biasing portion **32**. The cam portion **44** is provided at a position including the turning axis of the movable member **40** as viewed in the connector width direction, and is formed such that the sectional shape thereof at a right angle to the turning axis is a substantially rectangular shape.
(63) When the movable member **40** is at the closed position, the cam portion **44** contacts a lower surface of the biasing portion **32** at a first restriction target surface **44**A forming a flat upper surface at the closed position (FIGS. **6**C and **7**C), and as a result, the movable member **40** is reliably maintained at the closed position. When the movable member **40** is at the open position, the cam portion **44** contacts the lower surface of the biasing portion **32** at a second restriction target surface

44B forming a flat upper surface at the open position (see FIG. **8**C), and as a result, the movable member **40** is reliably maintained at the open position.
(64) In the present embodiment, in a state in which the cam portion **44** contacts the lower surface of the biasing portion **32** at the closed position or the open position, the movable arm portion **31**B of the metal fitting **30** is not elastically deformed, i.e., biasing force from the biasing portion **32** does not act on the cam portion **44**. However, as a modification, at least at one of the closed position or the open position, in a state in which the movable arm portion **31**B of the metal fitting **30** is slightly elastically deformed, the biasing portion **32** may contact the cam portion **44** such that moderate biasing force from the biasing portion **32** acts on the cam portion **44**. As another modification, at least at one of the closed position or the open position, a clearance may be formed in the up-down direction between the cam portion **44** and the biasing portion **32**.
(65) A cam surface **44**C is formed between the first restriction target surface **44**A and the second restriction target surface **44**B at an outer peripheral surface (outer peripheral surface parallel with the connector width direction) of the cam portion **44**. The cam surface **44**C is a protruding curved surface formed at one corner portion of the cam portion **44**. In the present embodiment, in a course of the movable member **40** turning between the closed position and the open position, the cam portion **44** presses the biasing portion **32** upward by the cam surface **44**C, and accordingly, elastically displaces the movable arm portion **31**B. Then, the cam portion **44** receives reactive force, i.e., downward biasing force, from the biasing portion **32**.
(66) The connector **1** configured as described above is assembled in the following manner. First, the terminals **20** are press-fitted in and attached to the terminal housing portions **19** of the housing **10** from the front. Moreover, the movable member **40** held in the posture at the closed position is attached to the housing **10** from above. At this point, the side plate portions **43** of the movable member **40** are housed in the side plate housing portions **10**A, and the cam portions **44** are housed in the cam housing portions **10**B. Next, the metal fittings **30** are press-fitted in and attached to the metal fitting housing portions **17**A of the housing **10** from the back. At this point, the locking leg portions **34** are fitted in the locking target end portions **17**B, and are locked to the locking target end portions **17**B from below. Moreover, the biasing portions **32** of the metal fittings **30** are positioned immediately above the cam portions **44**, and accordingly, upward movement of the cam portions **44** is restricted. The connector **1** is completed in such a manner that the terminals **20**, the metal fittings **30**, and the movable member **40** are attached to the housing **10** as described above.
(67) Note that in the present embodiment, the terminals **20**, the movable member **40**, and the metal fittings **30** are attached to the housing **10** in this order, but a step of attaching the terminals **20** may be performed after a step of attaching the metal fittings **30** and the movable member **40** or may be performed simultaneously. In the present embodiment, the movable member **40** in the posture at the closed position is attached to the housing **10**, but the posture of the movable member **40** upon attachment thereof is not limited to above and may be, e.g., the posture at the open position.
(68) Next, an operation of inserting the flat conductor C into the connector **1** or removing the flat conductor C from the connector **1** will be described.
(69) First, the connection portions **24** of the terminals **20** of the connector **1** are soldered to the corresponding circuit portions of the circuit board (not shown), and the fixing leg portions **33** of the metal fittings **30** are soldered to the corresponding portions of the circuit board. By such soldering of the connection portions **24** and the fixing leg portions **33**, the connector **1** is attached to the circuit board.
(70) Next, as shown in FIGS. **6**A to **6**C, at the back of the connector **1** in a state in which the movable member **40** is at the closed position, the flat conductor C is positioned so as to extend in the front-back direction (X-axis direction) along the mount surface of the circuit board (not shown) (also see FIG. **1**). Next, the flat conductor C is inserted forward (X1 direction) into the receiving space **11** of the connector **1**.
(71) In the present embodiment, the second elastic portion **21**B-**2** of the terminal **20** having the

lower contact portion **21**B-**3** is bent at the back end of the first elastic portion **21**B-**1**, and extends forward. Thus, when the flat conductor C is inserted into the receiving space **11** from back, the flat conductor C is smoothly guided forward by back end portions of the elastic arm portions **21**B, i.e., bent portions of the elastic arm portions **21**B curved in a backwardly-raised shape, and reaches the position of the lower contact portions **21**B-**3**. Thus, at any portion, the terminal **20** is not buckled due to contact with the front end of the flat conductor C. As a result, the terminals **20** and the flat conductor C can properly contact each other.

(72) In a course of the flat conductor C being inserted into the receiving space **11**, the front end of the flat conductor C contacts the lower contact portions **21**B-**3** of the second elastic portions **21**B-**2** of the terminals **20** and the upper contact portions **22**A of the upper arm portions **22** of the terminals **20**. Then, the front end of the flat conductor C pushes down the second elastic portions **21**B-**2** to elastically displace the second elastic portions **21**B-**2** downward, and pushes up the upper arm portions **22** to elastically displace the upper arm portions **22** upward. As a result, a portion between the lower contact portions **21**B-**3** and the upper contact portions **22**A is expanded. At this point, the first elastic portions **21**B-**1** are also elastically displaced downward in response to elastic displacement of the second elastic portions **21**B-**2**.

(73) At the position of each locking portion **42** of the movable member **40** in the connector width direction, the front end of the flat conductor C contacts the guide surface **42**A of the locking portion **42** to push up the locking portion **42**. Since the locking portions **42** are pushed up, the entirety of the movable member **40** is moved upward. Accordingly, the biasing portions **32** of the metal fittings **30** are pushed up by the cam portions **44**, and the movable arm portions **31**B of the metal fittings **30** are elastically displaced upward. That is, elastic displacement of the movable arm portions **31**B allows upward movement of the locking portions **42**.

(74) Since the portion between the lower contact portions **21**B-**3** and the upper contact portions **22**A of the terminals **20** is expanded and the locking portions **42** of the movable member **40** are moved upward as described above, the flat conductor C can be inserted further forward. As shown in FIGS. **7**A to **7**C, the flat conductor C is inserted until contacting the front wall **15** (see FIGS. **7**A and **7**B) of the housing **10**. At this point, the shoulder portions C3 of the flat conductor C are positioned close to the back end wall portions **18** of the housing **10** at the back thereof, as shown in FIG. **2**.

(75) In a state in which insertion of the flat conductor C is completed, the elastically-displaced state of the elastic arm portions **21**B (first elastic portions **21**B-**1** and second elastic portions **21**B-**2**) of the lower arm portions **21** and the upper arm portions **22** is maintained as shown in FIG. **7**A, and the flat conductor C is sandwiched by the lower contact portions **21**B-**3** and the upper contact portions **22**A. That is, the upper contact portions **22**A press the flat conductor C from above while the lower contact portions **21**B-**3** contact the circuit portions of the flat conductor C with contact pressure. In this manner, the electrical conduction state of the terminals **20** and the flat conductor C is maintained.

(76) In a course of inserting the flat conductor C, when the locking portions **42** reach the position of the cutouts C1 after the ear portions C2 of the flat conductor C have passed through the position of the locking portions **42**, the movable member **40** returns to the closed position, and the locking portions **42** enter the cutouts C1 from above as shown in FIG. **7**B. As a result, the locking surfaces **42**B of the locking portions **42** are positioned lockable to the locking target portions C2A of the flat conductor C from the back, and accordingly, unexpected detachment of the flat conductor C is prevented. As shown in FIG. **7**C, the lower surface (rolled surface) of the biasing portion **32** of the metal fitting **30** contacts the first restriction target surface **44**A of the cam portion **44** of the movable member **40**, and the movable member **40** is maintained at the closed position. In this manner, an operation of connecting the flat conductor C to the connector **1** is completed.

(77) In the state shown in FIGS. **7**A to **7**C, i.e., when the flat conductor C connected to the connector **1** is purposefully removed from the connector **1**, the movable member **40** at the closed

position is turned to the open position shown in FIGS. **8**A to **8**C. In a course of the movable member **40** turning to the open position, the cam surfaces **44**C of the cam portions **44** press the biasing portions **32** of the metal fittings **30** upward while slidably contacting the lower surfaces of the biasing portions **32**. When the biasing portions **32** are pressed, the movable arm portions **31**B of the metal fittings **30** are elastically displaced upward, and accordingly, the biasing portions **32** are displaced upward. That is, the cam portions **44** are allowed to be further turned while receiving the biasing force from the biasing portions **32**.

(78) In the present embodiment, the biasing portion **32** biases the cam portion **44** at the plate surface (rolled surface) thereof. Thus, as compared to a case where a plate-thickness surface (fracture surface) of the biasing portion biases the cam portion, the biasing portion **32** can contact the cam portion **44** at a smoother surface, and the area of contact between the biasing portion **32** and the cam portion **44** can be increased. As a result, even if the biasing portion **32** and the cam portion **44** repeatedly slidably contact each other due to movement of the movable member **40**, abrasion of the cam portion **44** is less likely to be caused, and lowering of the biasing force of the biasing portion **32** can be favorably avoided.

(79) When the movable member **40** reaches the open position, the second restriction target surface **44**B of the cam portion **44** forms the upper surface, and the movable arm portion **31**B of the metal fitting **30** is no longer elastically displaced and returns to a free state, as shown in FIG. **8**C. As a result, the lower surface of the biasing portion **32** contacts the second restriction target surface **44**B of the cam portion **44**, and the movable member **40** is maintained at the open position. Since the movable member **40** is maintained at the open position as described above, unexpected movement of the movable member **40** to the closed position can be avoided.

(80) As a result of movement of the movable member **40** to the open position, the locking portion **42** of the movable member **40** is separated upward from the cutout C1 of the flat conductor C as shown in FIG. **8**B, and the flat conductor C is allowed to be removed. Moreover, this state is maintained by contact between the lower surface of the biasing portion **32** and the second restriction target surface **44**B of the cam portion **44** as shown in FIG. **8**C. The flat conductor C is pulled backward (X2 direction), and in this manner, the flat conductor C is removed from the connector **1** without difficulty and the removal operation is completed.

(81) In the present embodiment, the fixing leg portion **33** is provided at the metal fitting **30** and is soldered to the mount surface of the circuit board, and therefore, when the biasing portion **32** receives upward force from the cam portion **44** of the movable member **40** in a course of turning the movable member **40**, the soldered portion of the fixing leg portion **33** can provide resistance against the upward force, and as a result, detachment of the metal fitting **30** from the housing can be avoided.

(82) In the present embodiment, the locking leg portion **34** is provided at the metal fitting **30**, and therefore, detachment of the metal fitting **30** from the housing **10** can be avoided when the biasing portion **32** receives the upward force from the cam portion **44** of the movable member **40** because the locking leg portion **34** is locked to the locking target end portion **17**B of the housing **10** from below. Specifically, the fixing leg portion **33** is soldered to the mount surface of the circuit board yet before the connector **1** is mounted on the circuit board, and therefore, the soldered portion of the fixing leg portion **33** cannot provide resistance against the upward force. Thus, a technique of providing the locking leg portion **34** to provide resistance against the upward force by locking force between the locking leg portion **34** and the locking target end portion **17**B is extremely effective before connector mounting.

(83) In the present embodiment, the circuit portions of the flat conductor C are exposed at the lower surface of the flat conductor C. Instead, the circuit portions may be exposed at the upper surface of the flat conductor C. In this case, the upper arm portion **22** of the terminal **20** contacts the circuit portion at the upper contact portion **22**A. Alternatively, the circuit portions may be exposed at both the lower and upper surfaces of the flat conductor C. In this case, the lower contact portion **21**B-**3**

contacts the lower circuit portion, and the upper contact portion **22**A contacts the upper circuit portion.
(84) In the present embodiment, the biasing portion **32** of the metal fitting **30** biases the cam portion **44** of the movable member **40**, but the movable member portion to be biased is not necessarily the cam portion and may be other portions of the movable member.
(85) The foregoing detailed description has been presented for the purposes of illustration and description. Many modifications and variations are possible in light of the above teaching. It is not intended to be exhaustive or to limit the subject matter described herein to the precise form disclosed. Although the subject matter has been described in language specific to structural features and/or methodological acts, it is to be understood that the subject matter defined in the appended claims is not necessarily limited to the specific features or acts described above. Rather, the specific features and acts described above are disclosed as example forms of implementing the claims appended hereto.

Claims

1. A flat conductor electric connector to which a flat conductor extending in a front-back direction is connected, comprising: a housing formed with a receiving space which is opened to a back such that the flat conductor is inserted forward into the housing; and multiple terminals arrayed and held on the housing in a state in which plate surfaces of the terminals are at a right angle to a terminal array direction which is a direction at a right angle to both the front-back direction and a thickness direction of the flat conductor, wherein each terminal has an arm portion extending in the front-back direction, the arm portion has a support target arm portion supported on an inner surface of a wall portion and extending in the front-back direction, and an elastic arm portion extending backward from the support target arm portion, the elastic arm portion has a first elastic portion extending from a back end of the support target arm portion and elastically displaceable in the thickness direction of the flat conductor without the support target arm portion supported on the wall portion, and a second elastic portion positioned on a receiving space side with respect to the first elastic portion, bent forward from a back end of the first elastic portion, and elastically displaceable in the thickness direction of the flat conductor, and the second elastic portion has a contact portion contactable with the flat conductor with contact pressure.
2. The flat conductor electric connector according to claim 1, wherein the first elastic portion is formed thicker and longer than the second elastic portion as viewed in the terminal array direction, and a front end of the first elastic portion is positioned at a front with respect to a front end of the second elastic portion.
3. The flat conductor electric connector according to claim 1, wherein each terminal has another arm portion positioned on an opposite side of the receiving space from the arm portion and extending in the front-back direction, the other arm portion has another contact portion contacting the flat conductor with contact pressure to sandwich the flat conductor in cooperation with the contact portion, and a distance between the contact portion and the other contact portion in the front-back direction is smaller than a thickness dimension of the flat conductor.
4. The flat conductor electric connector according to claim 1, wherein each terminal has another arm portion positioned on an opposite side of the receiving space from the arm portion and extending in the front-back direction, the other arm portion has another contact portion contacting the flat conductor with contact pressure to sandwich the flat conductor in cooperation with the contact portion, and a distance between the contact portion and the other contact portion in the front-back direction is smaller than twice as great as a clearance dimension between the contact portion and the other contact portion in the thickness direction of the flat conductor.